

Scenario

“Professor Kılıç” from the Molecular Biology department has sent you a sequencing data file from a bacterial isolate. The isolate was recovered from a patient whose infection is not responding to standard antibiotics, and the lab sequenced it using Oxford Nanopore (ONT). At this point the lab does not know which organism it is, and the clinical team needs to know what it is resistant to. Professor Kılıç is not a technical person, so he needs a clear, practical answer.

Objective

Using only the raw Nanopore reads, determine two things: (1) what the organism is, and (2) what antimicrobial resistance it carries. Then report your findings clearly. The approach is entirely up to you.

Part 1: The Analysis

You are given a single file, `unknown_isolate.fastq.gz`, containing the raw ONT reads. Nothing else is provided: the organism, the quality of the data, and the details of the run are all unknown, and working them out is part of the task. There are no fixed steps and no required tools. You decide what to run, in what order, and how to confirm that your answer is reliable. Choose your own approach, and be prepared to justify it.

Questions worth answering:

- What is the organism? Identify the species, and its strain type or lineage where possible.
- Which resistance genes does it carry, and is there anything a clinician should be concerned about, such as resistance to last-line antibiotics?
- Where does the resistance sit: on the chromosome, or on a plasmid that could spread to other bacteria?
- Is there anything else worth noting, such as virulence factors?

Part 2: The Report

Prepare a `findings.md` file describing what you found and what it means, written so that both Professor Kılıç (non-technical) and a fellow bioinformatician can follow it. Also record the tools and reference-database versions you used, since resistance databases are updated regularly and the same reads can yield different results across versions.

Part 3: What to Submit

Submit your work as a single ZIP file. (A link to a public GitHub repository is also acceptable, but not required.) It should contain: findings.md, your report; code/, a folder holding every command, script, or workflow you ran, in any format (shell, a notebook, Snakemake, Nextflow, or whatever you prefer; no framework is required); and a short README explaining how to run it.

Part 4: Communication

Include a short note to “Professor Kılıç” within your report. In plain terms, explain what the organism is,

what it is resistant to, what that means for the patient, and what you would recommend as next steps.

What We Look For

- Correct conclusions: the right organism, and an accurate account of its resistance.
- A method we can reproduce from your code and notes.
- A realistic sense of certainty: be clear about what the data does and does not show.
- Clear communication: a report that a clinician and a bioinformatician can both use.

Working Notes

You may use any tools, databases, and online resources you wish; there are no restrictions. Work at your own pace, and if anything here is unclear, please ask before you begin.

Example Data: [unknown_isolate.fastq.gz](#)